FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2019

PAPER – 2

TEST DATE: 28-04-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

Sol. $v\cos\theta = v_0$

$v = v_0\sec\theta$ …(i)

at $\theta = 60°$,

$v = 2\sec 60° = 2\times 2 = 4$ m/s

$v = v_0\sec\theta$

$\frac{dv}{dt} = v_0\sec\theta\tan\theta\frac{d\theta}{dt}$

$a = v_0\sec\theta\tan\theta\frac{v\sin\theta}{\ell}$

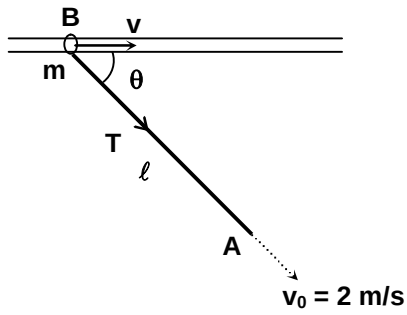
$a = \frac{v_0^2}{\ell}\tan^2\theta\sec\theta$

$T\cos\theta = ma$

$T = ma\sec\theta = \frac{mv_0^2}{\ell}\tan^2\theta\sec^2\theta$

$T = \frac{0.5\times 4\times 3\times 4}{1} = 24$ N

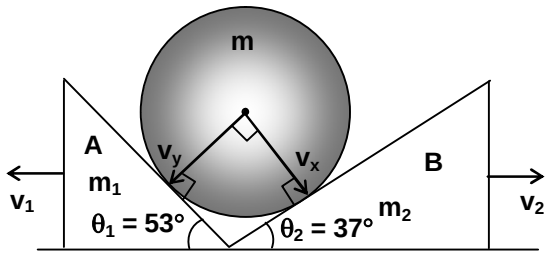
T = 24 N



2. A, D

Sol. $v_x = v_2 \sin\theta_2 = v_2 \sin 37° = \frac{3}{5}v_2$ ...(i)

$v_y = v_1 \sin\theta_1 = v_1 \sin 53° = \frac{4}{5}v_1$ ...(ii)



using conservation of momentum of system along horizontal direction

$m_2v_2 - m_1v_1 + m(v_x \sin\theta_2 - v_y \sin\theta_1) = 0$

$16v_2 - 9v_1 + 25\left(\frac{9}{25}v_2 - \frac{16}{25}v_1\right) = 0$

$25v_2 = 25\, v_1$

$\therefore\ v_1 = v_2$

Velocity of cylinder, $v = \sqrt{v_x^2 + v_y^2} = \frac{v_1}{5}\sqrt{(3)^2 + (4)^2}$

$\therefore\ v = v_1 = v_2$ ...(iii)

Now, using conservation of energy of the system

$mgh = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}mv^2$

$mgh = \frac{1}{2}(m_1 + m_2 + m)v^2$

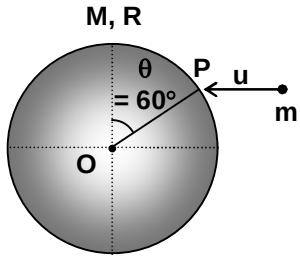
$25gh = \frac{1}{2} \times 50v^2$

$\Rightarrow\ v = \sqrt{gh} = \sqrt{10 \times 0.4} = 2$ m/s

$\therefore\ v = v_1 = v_2 = 2$ m/s

3. B, C

Sol. Using conservation of angular momentum of the system about a horizontal axis passing through centre O.



$muR\cos\theta = \left(\frac{MR^2}{2} + mR^2\right)\omega_0$

$muR\cos 60° = \left(\frac{M}{2} + m\right)R^2\omega_0$

$0.5 \times 20 \times \frac{1}{2} = 2.5 \times \frac{1}{4} \times \omega_0$

$\Rightarrow\ \omega_0 = 8$ rad/s

Now, using conservation of energy of the system

$mgR(1 - \cos\theta) = \frac{1}{2}\left(\frac{MR^2}{2} + mR^2\right)\left(\omega_0^2 - \omega^2\right)$

$mg\frac{R}{2} = \frac{1}{2}\left(\frac{M}{2} + m\right)R^2\left(\omega_0^2 - \omega^2\right)$

$0.5 \times \frac{10}{2} = \frac{5}{4} \times \frac{1}{4}\left(\omega_0^2 - \omega^2\right)$

$\Rightarrow\ \omega_0^2 - \omega^2 = 8$

$\omega^2 = \omega_0^2 - 8 = 64 - 8 = 56$

$\omega = \sqrt{56}$ rad/s

Now, $(M + m)g - F = (M + m)\frac{\omega^2 R}{9}$

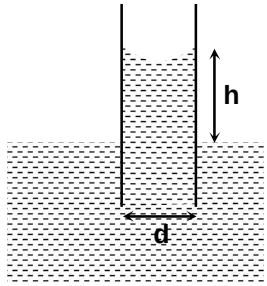
$45 - F = \frac{9}{2} \times 56 \times \frac{0.25}{9}$

$\therefore$ F = 38 N (vertically upward)

4. A, D

Sol. $2T\ell = \rho d\ell hg$

$$h = \frac{2T}{\rho g d} = \frac{2 \times 7 \times 10^{-2}}{10^3 \times 10 \times 1 \times 10^{-4}} = 14 \times 10^{-2} m = 14 cm$$



Force of mutual attraction between the plates is

$$F = \int_0^h \rho g y \ell dy = \frac{\rho g \ell h^2}{2}$$

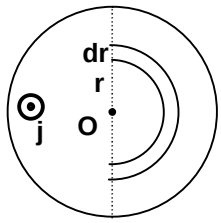
$$= \frac{10^3 \times 10 \times 0.10 \times (14 \times 10^{-2})^2}{2} = 19.6/2 = 9.8 \text{ N}$$

5. B, C

Sol. When $0 \le r \le R$

$B2\pi r = \mu_0 j \pi r^2$

$$B = \frac{1}{2}\mu_0 jr$$



Magnetic field at r = R/2, $B = \frac{\mu_0 jR}{4}$

The linear current density i = jdr

The force of interaction per unit length between two halves is

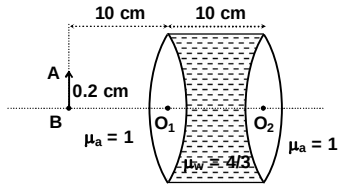
$$F = \int_0^R iB2r = \int_0^R jdr\left(\frac{\mu_0 jr}{2}\right)2r$$

$$= \mu_0 j^2 \int_0^R r^2 dr$$

$$F = \left(\frac{\mu_0 j^2 R^3}{3}\right)$$

6. A, C

Sol. For the image formed by the first lens



$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \left(\frac{\mu_2 - \mu_1}{R_1}\right) + \left(\frac{\mu_3 - \mu_2}{R_2}\right)$$

$$\Rightarrow \frac{4}{3v} - \frac{1}{(-10)} = \frac{\left(\frac{3}{2} - 1\right)}{10} + \frac{\left(\frac{4}{3} - \frac{3}{2}\right)}{-10}$$

$$\frac{4}{3v} + \frac{1}{10} = \frac{1}{20} + \frac{1}{60}$$

$$\Rightarrow \frac{4}{3v} = \frac{1}{15} - \frac{1}{10}$$

$\Rightarrow$ v = –40 cm

Lateral magnification, $m_1 = \frac{\mu_1}{\mu_3}\frac{v}{u} = \frac{1 \times 3}{4}\left(\frac{-40}{-10}\right) = +3$

For the image formed by the second lens

$$\frac{\mu_3}{v}-\frac{\mu_1}{u}=\left(\frac{\mu_2-\mu_1}{R_1}\right)+\left(\frac{\mu_3-\mu_2}{R_2}\right)$$

$$\Rightarrow \frac{1}{v}-\frac{4}{3(-50)}=\left(\frac{\frac{3}{2}-\frac{4}{3}}{10}\right)+\left(\frac{1-\frac{3}{2}}{-10}\right)$$

$$\Rightarrow \frac{1}{v}+\frac{2}{75}=\frac{1}{60}+\frac{1}{20}$$

$$\Rightarrow \frac{1}{v}=\frac{1}{15}-\frac{2}{75}$$

$\Rightarrow v = +25$ cm

The position of final image formed is 25 cm right to the second lens. Lateral magnification,

$$m_2=\frac{\mu_1}{\mu_3}\frac{v}{u}=\frac{4}{3\times 1}\left(\frac{25}{-50}\right)=-\frac{2}{3}$$

Overall magnification, $m = m_1 \times m_2 = (+3) \times \left(-\frac{2}{3}\right) = -2$

The size of the final image formed = 2×0.2 cm = 0.4 cm

7. B

Sol. (P) $x_1 + 2x_2 + 2x_3 + 2x_4 + 2x_5 = x$

$$\frac{F}{k_1}+\frac{4F}{k_2}+\frac{4F}{k_3}+\frac{4F}{k_4}+\frac{4F}{k_5}=x$$

$$\frac{F}{k}+\frac{4F}{k}+\frac{4F}{k}+\frac{4F}{2k}+\frac{4F}{2k}=x$$

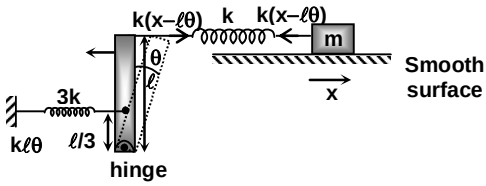
$$\frac{13F}{k}=x \Rightarrow F=\frac{kx}{13} \Rightarrow \frac{d^2x}{dt^2}=-\frac{kx}{13m}$$

$$T=2\pi\sqrt{\frac{13m}{k}}$$

(Q) $$m\frac{d^2x}{dt^2}=-8kx \Rightarrow \frac{d^2x}{dt^2}=-\left(\frac{8k}{m}\right)x$$

$$\therefore\ T=2\pi\sqrt{\frac{m}{8k}}=\pi\sqrt{\frac{m}{2k}}$$

(R) Since rod is light



$$3\frac{k\ell\theta}{3}\frac{\ell}{3}=K(x-\ell\theta)\ell$$

$$\frac{k\ell}{3}\theta=kx-k\ell\theta$$

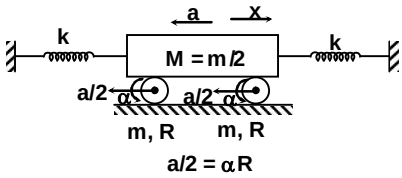
$$\theta=\frac{3x}{4\ell}$$

$$m\frac{d^2x}{dt^2}=-k(x-\ell\theta)=-k\left(x-\frac{3x}{4}\right)$$

$$\frac{d^2x}{dt^2}=-\frac{kx}{4m}$$

$$\Rightarrow T=2\pi\sqrt{\frac{4m}{k}}=4\pi\sqrt{\frac{m}{k}}$$

(S) $2kx - 2f_1 = Ma = \frac{ma}{2}$ …(i)



$f_1 + f_2 = \frac{ma}{2}$ …(ii)

$(f_1 - f_2)R = \frac{mR^2}{2}\alpha$

$(f_1 - f_2) = \frac{m\alpha R}{2} = \frac{ma}{4}$ …(iii)

Adding (i), (ii) and (iii), we get

$2kx = \frac{5ma}{4} \Rightarrow a = \frac{8kx}{5m}$

$\frac{d^2x}{dt^2} = -\left(\frac{8k}{5m}\right)x \Rightarrow T = 2\pi\sqrt{\frac{5m}{8k}} = \pi\sqrt{\frac{5m}{2k}}$

8. D

Sol. (P) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 = 10 : 80 : 10 = 0.5 : 4 : 0.5$

$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4$ m

$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2\times10^{11}}{8000}} = 5000$ m/s

$f = \frac{v}{\lambda} = \frac{5000}{0.4} = 12500 = 12.5$ kHz '

(Q) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 = 30 : 40 : 30 = 1.5 : 2 : 1.5$

$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4$ m

$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6\times10^{11}}{2500}} = 8000$ m/s

$f = \frac{v}{\lambda} = \frac{8000}{0.4} = 20000 = 20$ kHz

(R) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 = 10 : 60 : 30 = 0.5 : 3 : 1.5$

$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4$ m

$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2\times10^{11}}{8000}} = 5000$ m/s

$f = \frac{v}{\lambda} = \frac{5000}{0.4} = 12500 = 12.5$ kHz

(S) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3 = 15 : 60 : 25 = 1.5 : 6 : 2.5$

$\Rightarrow \frac{10\lambda}{2} = 1 \Rightarrow \lambda = 0.2$ m

$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6\times10^{11}}{2500}} = 8000$ m/s

$f = \frac{v}{\lambda} = \frac{8000}{0.2} = 40000 = 40$ kHz

9. C

Sol. $\Delta Q_{AB} = nC_V\Delta T = 2\times\frac{3R}{2}\times 300 = 900\ R$

$\Delta W_{AB} = 0$

$\Delta Q_{BC} = nC_P\Delta T = 2\times\frac{5R}{2}\times(1800-600) = 6000\ R$

$\Delta W_{BC} = P\Delta V = nR\Delta T = 2R(1800-600) = 2400\ R$

$\Delta Q_{CD} = nC_V\Delta T = 2\times\frac{3R}{2}\times(900-1800) = -2700\ R$

$\Delta W_{CD} = 0$

$\Delta Q_{DA} = nC_P\Delta T = 2\times\frac{5R}{2}\times(300-900) = -3000\ R$

$\Delta W_{DA} = P\Delta V = nR\Delta T = 2R(300-900) = -1200\ R$

10. A

Sol. (P) $z = R + (X_L - X_C)i = 3 + (8-4)i = 3+4i$

$|z| = \sqrt{(3)^2+(4)^2} = 5\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{100}{5} = 20\ A$

(Q) $\frac{1}{z} = \frac{1}{X_L i} + \frac{1}{-X_C i} + \frac{1}{R} = \frac{1}{6i} - \frac{1}{3i} + \frac{1}{8} = \left(\frac{1}{3}-\frac{1}{6}\right)i + \frac{1}{8} = \frac{1}{8} + \frac{1}{6}i$

$\Rightarrow \quad z = \frac{48}{(6+8i)}$

$\Rightarrow \quad z = \frac{48(6-8i)}{100}$

$\Rightarrow \quad |z| = \frac{48}{100}\times 10$

$\Rightarrow |z| = 4.8\ \Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{120}{4.8} = 25\ A$

(R) $\frac{1}{z'} = \frac{1}{X_L i} + \frac{1}{-X_C i} = \frac{1}{6i} - \frac{1}{3i} = i\left(\frac{1}{3}-\frac{1}{6}\right) = \frac{i}{6}$

$z' = \frac{6}{i} = -6i$

$z = R + z' = 8 - 6i$

$|z| = \sqrt{(8)^2+(6)^2} = 10\,\Omega$

$I_{rms} = \frac{\varepsilon_{rms}}{|z|} = \frac{50}{10} = 5\ A$

(S) $z_1 = R_1 + X_L i = 8 + 6i \Rightarrow I_{1(rms)} = \frac{\varepsilon_{rms}}{|z_1|} = \frac{100}{10} = 10\ A$

$z_2 = R_2 - X_C i = 3 - 4i \Rightarrow I_{2(rms)} = \frac{\varepsilon_{rms}}{|z_2|} = \frac{100}{5} = 20\ A$

$\therefore\ I_{rms} = \sqrt{I_{1(rms)}^2 + I_{2(rms)}^2} = \sqrt{(10)^2+(20)^2} = \sqrt{500} = 10\sqrt{5}\ A$

SECTION – D

11. 00057.76

Sol. The frequency of vibration of air column in the pipe

$$f_0 = \frac{3v}{4\ell_0} = \frac{3\times 340}{4\times 0.85} = 300\ \text{Hz}$$

$$f_s - f_0 = 4$$

$$f_s = f_0 + 4 = 300 + 4$$

$$\frac{4}{2\ell}\sqrt{\frac{F}{\mu}} = 304$$

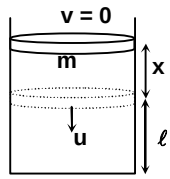
$$\frac{4}{2\times 0.5}\sqrt{\frac{F\times 0.5}{5\times 10^{-3}}} = 304$$

$$\sqrt{\frac{F}{10^{-2}}} = 76$$

$\therefore$ F = 57.76 (Newton)

12. 00001.28

Sol. Let the vertical upward displacement of the piston before coming to a complete stop is 'x'.



In initial and final equilibrium of the piston

$$PA = \frac{nRT_0}{\ell} = \frac{nRT}{(\ell + x)} = mg$$

$nRT_0 = mg\ell$ …(i)

$nRT = mg(\ell + x)$ …(ii)

Now, $nR\Delta T = mgx$ …(iii)

$$\frac{1}{2}mu^2 = nC_v\Delta T + mgx$$

$$\frac{1}{2}mu^2 = \frac{n3R}{2}(T - T_0) + mgx$$

$$\frac{1}{2}mu^2 = \frac{3nR\Delta T}{2} + mgx$$

$$\frac{1}{2}mu^2 = \frac{5}{2}mgx$$

$$x = \frac{u^2}{5g} = \frac{8\times 8}{5\times 10} = 1.28\ \text{meter}$$

$\therefore$ vertical upward displacement of the piston = 1.28 meter.

13. 00002.84

Sol. Since the central maxima is formed at point 'P'

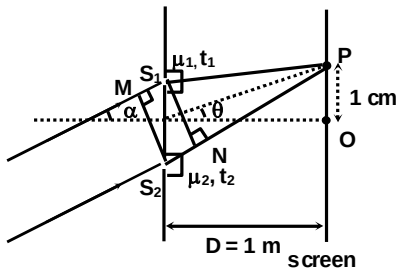
d sin α + $(\mu_1 - 1)t_1 = (\mu_2 - 1)t_2$ + d sin θ

$$2\times 10^{-3}\times \sin 30° + (1.5-1)t_1 = (1.8-1)3\times 10^{-3} + 2\times 10^{-3}\times \frac{1}{100}$$

$\Rightarrow 1 \times 10^{-3} + 0.5t_1 = 2.4 \times 10^{-3} + 0.02 \times 10^{-3}$

$\Rightarrow t_1 = 2.84 \times 10^{-3}$

$\Rightarrow t_1 = 2.84$ mm



14. 00003.98

Sol. Energy of each photon of light incident on the metal surface

$$E = \frac{hC}{\lambda} = \frac{1240}{500} = 2.48 \text{ eV}$$

Kinetic energy of fastest photoelectron emitted

$K_{max} = E - \phi = 2.48 - 1.90 = 0.58$ eV

$$\therefore E_n = -13.6\frac{z^2}{n^2}$$

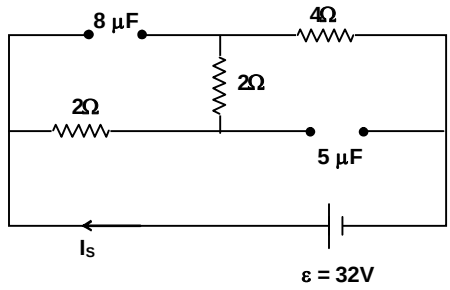
$$E_4 = -13.6 \times \frac{(2)^2}{4 \times 4} = -3.40 \text{ eV}$$

Energy of the photon emitted during combination

$\Delta E = K_{max} - E_4 = 0.58 - (-3.40) = 3.98$ eV

15. 00001.44

Sol. The equivalent circuit in steady state is shown.



The current through the battery in steady state is

$$I_s = \frac{32}{8} = 4A$$

Potential drop across 5 μF capacitor in steady state,

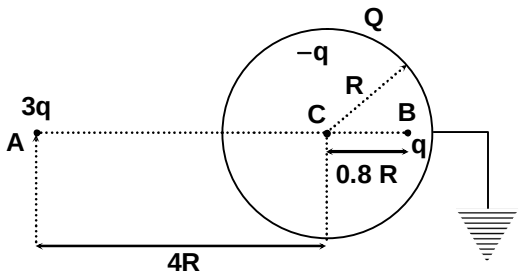
$V_0 = 4\ (2 + 4) = 24$ Volt

$\therefore$ Energy stored in 5 μF capacitor in steady state

$$U = \frac{1}{2} \times 5 \times 10^{-6} \times (24)^2 = 1.44 \text{ milli Joule}$$

16. 00007.50

Sol. Let the charge on the conducting shell after being earthed is Q.



$$\frac{KQ}{R} + \frac{3kq}{4R} = 0$$

$$Q = \frac{-3q}{4}$$

$\therefore$ Total charge that will flow to the earth is

$$\Delta Q = 3q - \left(\frac{-3q}{4}\right) = \frac{15q}{4} = \frac{15}{4} \times 2 = 7.50 \ \mu C$$

17. 00002.70

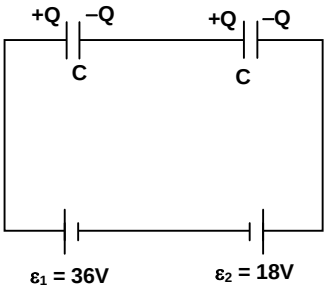
Sol. Charge on each capacitor, when switch S is opened

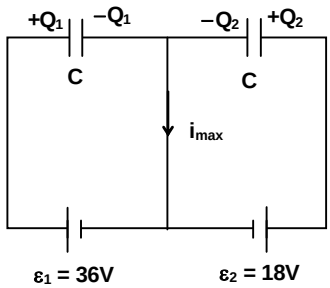
$$Q = \frac{C}{2}(\varepsilon_1 - \varepsilon_2) = \frac{10}{2} \times 18 = 90 \ \mu C$$

Charge on the capacitor when the current through the inductor is maximum after closing the switch 'S'.

$Q_1 = C\varepsilon_1 = 10 \times 36 = 360 \ \mu C$

$Q_2 = C\varepsilon_2 = 10 \times 18 = 180 \ \mu C$





Now, $\Delta Q_1\varepsilon_1 + \Delta Q_2\varepsilon_2 = \left(\frac{1}{2}C\varepsilon_1^2 + \frac{1}{2}C\varepsilon_2^2\right) - \frac{1}{2}\times\frac{C}{2}\left(\varepsilon_1 - \varepsilon_2\right)^2 + \frac{1}{2}Li_{max}^2$

$10^{-6}\left(270\times 36\right) + 10^{-6}(270\times 18)$

$= 10^{-6}\left\{\frac{1}{2}\times 10\times(36)^2 + \frac{1}{2}\times 10\times(18)^2 - \frac{1}{2}\times 5\times(18)^2\right\} + \frac{1}{2}\times 2\times 10^{-3} i_{max}^2$

$14580 = (8100 - 810) + 1\times 10^3\times i_{max}^2$

$10^3\times i_{max}^2 = 7290$

$i_{max} = \sqrt{7.29} = 2.70$ ampere

18. 00052.80

Sol. Thermal resistance of the cylinders are

$R_1 = \int_a^{2a}\frac{dr}{2\pi K_1 r\ell} = \frac{1}{2\pi K_1\ell}\ell n2$

$R_2 = \int_{2a}^{4a}\frac{dr}{2\pi K_2 r\ell} = \frac{1}{2\pi K_2\ell}\ell n2$

Heat current flowing between the innermost and outermost cylindrical walls is

$i = \frac{\theta_1 - \theta_2}{R_1 + R_2}$

$\therefore\ \theta_1 - \theta = iR_1$

$\Rightarrow\ \theta_1 - \theta = \left(\frac{\theta_1 - \theta_2}{R_1 + R_2}\right)R_1$

$\Rightarrow\quad \theta_1 - \theta = \frac{(\theta_1 - \theta_2)\frac{1}{k_1}}{\left(\frac{1}{k_1} + \frac{1}{k_2}\right)}$

$\Rightarrow\quad \theta_1 - \theta = \frac{(\theta_1 - \theta_2)k_2}{(k_1 + k_2)}$

$\Rightarrow\quad \theta = \left(\frac{k_1\theta_1 + k_2\theta_2}{k_1 + k_2}\right) = \frac{(2k\times 84) + (3k\times 32)}{(2k + 3k)} = \frac{168 + 96}{5} = 52.8°C$

Chemistry PART – II

SECTION – A

19. B, D

20. A, B, C, D

Sol. Through SPM only migration of solvent particle is permissible and hence none will be able to make a mixture.

21. B, C, D

Sol. $SiCl_4 + 4H_2O \longrightarrow H_4SiO_4 + 4HCl$

$BiCl_3 + H_2O \longrightarrow BiOCl + 2HCl$

22. B, C

Sol. (A) $(SiH_3)_3N$ is a weaker base than $(CH_3)_3N$

(B) BF_3 is weaker Lewis acid than BI_3 due to back bonding in BF_3

(C) Solid N_2O_5 is ionic in nature. It exist as $NO_2^+NO_3^-$

(D) PH_3 is less basic than NH_3

23. A, B, C

Sol. $Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$

$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$

$Sn(OH)_2 + 2NaOH \longrightarrow Na_2SnO_2 + 2H_2O$

24. A, B, D

Sol.

	$3A(g) \rightarrow$	$2B(g) + 2C(g)$	
t = 0	P_0		
t = 20	$P_0 - x$	$\frac{2x}{3}$	$\frac{2x}{3}$
$t = \infty$	0	$\frac{2P_0}{3}$	$\frac{2P_0}{3}$

$\frac{4P_0}{3} = 4$ $P_0 = 3$ atm

$P_0 + \frac{x}{3} = 3.5$ $\Rightarrow$ $t_{50\%} = 20$ is the half life

$x = 1.5$

$t_{75\%} = 2 \times 20 = 40$ min

$t_{87.5\%} = 3 \times t_{50\%} = 3 \times 20 = 60$

$t_{99\%} = \frac{2}{3} \times t_{99.9\%}$

$= \frac{2}{3} \times 10 \times t_{50\%} = \frac{400}{3}$

25. A

Sol. $NiCl_4^{2-}$ sp^3 and two unpaired e^- – Tetrahedral

$Ni(CN)_4^{2-}$ dsp^2 and no unpaired e^- – Square planar

$Ni(CO)_4$ sp^3 and no unpaired e^- – Tetrahedral

$[Pd(NH_3)_4]^{2+}$ dsp^2 and no unpaired e^-. – Square planar.

26. D

Sol. (P) $Mg(OH)_2(s) \Leftrightarrow \underset{s}{Mg^{2+}} + \underset{2s}{2OH^-}$

$Ksp = 4s^3 = 4 \times 10^{-12}$

$s = 10^{-4}$

$[OH^-] = 2 \times 10^{-4}$

$\therefore$ pH = 10 + log 2

(Q) Basic buffer is formed. Hence,

pOH = $pk_b = \log\frac{1}{2}$ = 4 – log 2

pH = 10 + log 2

(R) A salt of weak acid and weak base is formed in which $k_a = k_b$

hence, pH = 7 or

pH = $\frac{1}{2}pk_w + \frac{1}{2}pk_a - \frac{1}{2}pk_b$

(S) $H_3PO_4 \Leftrightarrow H^+ + H_2PO_4^-$

$H_2PO_4^- \Leftrightarrow H^+ + HPO_4^{--}$

$HPO_4^{--} \Leftrightarrow H^+ + PO_4^{---}$

pH of 0.01 M Na_2HPO_4 = $\frac{pk_{a_2} + pk_{a_3}}{2} = \frac{8.102 + 12.5}{2}$

pH = $\frac{20.602}{2} = 10.301$

Hence, q and r.

27. A

Sol. Application of Le Chatelier's principle

28. B

SECTION – D

29. 00000.50

Sol. $\Delta T_f = iK_f m, \quad m = \frac{20}{172} \times \frac{1000}{50} = \frac{400}{172}$

$i = \frac{\Delta T_f}{K_f \times m} = \frac{2}{1.72 \times \frac{400}{172}}$

$i = 0.5$

30. 00001.09

Sol. $E^\circ_{cell} = -\frac{\Delta G^\circ}{nF}$

$= -\frac{(-2746) \times 1000}{26 \times 96500}$

$= 1.09\ V$

31. 00062.50

Sol. $K_c = \frac{Ka(HOCl)}{Ka(HCN)} = \frac{3 \times 10^{-8}}{4.8 \times 10^{-10}}$

$= 62.50$

32. 00001.81

Sol. $\frac{\Delta T_b}{\Delta T_f} = \frac{T_b^2 \times Lf}{T_f^2 \times L_v}$

$\frac{0.5}{\Delta T_f} = \frac{373 \times 373 \times 80}{273 \times 273 \times 540}$

$\Delta T_f = 1.81$

33. 03062.5

Sol. $K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$

Every 294 g $K_2Cr_2O_7$ gives 96 g of S, so 3062.5 kg of $K_2Cr_2O_7$ is required to produce 1000 kg of sulphur.

34. 00112.5

Sol. m. moles of $Cu^{2+} = 5$

m.moles of $H^+ = 100 \times 10^{-2} = 1$

m.moles of Cu^{2+} reduced $= \frac{1}{2} = 0.5$

m.moles of Cu^{2+} remaining $= 4.5$

$2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

n-Factor of $CuSO_4 = 1$

meq. of Cu^{2+} = m. mole of $Cu^{2+} = 4.5$

meq. of Cu^{2+} = meq. of $Na_2S_2O_3$

$4.5 = 0.04 \times V$

$V = 112.5$ ml.

35. 00001.33

Sol. $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$ at 127^0C the no of gaseous mole will remain same because H_2O will be in vapour form. These for the pressure change will be due to change in temperature

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{1}{300K} = \frac{P}{400K}$$

$$P = \frac{400}{300} = \frac{4}{3} = 1.33 atm$$

36. 00003.56

Sol. $pH = pk_{In} + \log\frac{[In^-]}{[InH]}$

$5 = pk_{In} + \log\frac{75}{25}$ $pk_{In} = 4.523$

$pH = 4.523 + \log\frac{10}{90} = 4.523 - .954 = 3.56$

Mathematics PART – III

SECTION – A

37. A, D

Sol. $x^2 - px - 1 = 0$ roots $x_1 < 0 < x_2$ (assume)

$x^2 - qx - 1 = 0$ roots $y_1 < 0 < y_2$

So (i) $x_1, x_2, y_1, y_2 \rightarrow$ not possible as $x_2 > 0$ and $y_1 < 0$

(ii) x_1, y_1, y_2, x_2 are in A.P. then $x_1 + x_2 = y_1 + y_2 \Rightarrow p = q$, not possible

(iii) x_1, y_1, x_2, y_2, then $x_2 - x_1 = y_2 - y_1 \Rightarrow \sqrt{p^2+4} = \sqrt{q^2+4}$

i.e. $p^2 = q^2 \Rightarrow p = -q \neq 0$

Also, $y_1 = \dfrac{x_1 + x_2}{2} = \dfrac{p}{2} \Rightarrow \dfrac{p^2}{2} - \dfrac{pq}{2} - 1 = 0$

Since $p = -q$

$\Rightarrow p^2 = \dfrac{4}{3}$; $p = \pm\dfrac{2}{\sqrt{3}}$

Hence, $(p, q) = \left(\dfrac{2}{\sqrt{3}}, \dfrac{-2}{\sqrt{3}}\right), \left(\dfrac{-2}{\sqrt{3}}, \dfrac{2}{\sqrt{3}}\right)$

38. B, D

Sol. Differentiate equation to get, $\dfrac{d^2y}{dx^2} = \dfrac{\sin x \cos^2 y + \sin y \cos^2 x}{\cos^3 y}$

Let $\sin x = t$ then $\sin y = 1 - t$. Also $\cos x = \sqrt{1-t^2}$ and $\cos y = \sqrt{1-\left(1-t^2\right)} = \sqrt{t(2-t)}$

Then, $\dfrac{d^2y}{dx^2} = t^{-3/2}\dfrac{1-t+t^2}{(2-t)^{3/2}}$

Since $\lim\limits_{x\to 0}\dfrac{t}{x} = 1$. So, $\alpha = \dfrac{3}{2}$ that gives $L = \lim\limits_{x\to 0} x^{\alpha}\dfrac{d^2y}{dx^2} = \dfrac{1}{2\sqrt{2}}$

39. B, C, D

Sol. Centre of circle will lie on perpendicular bisector of MN. Let the centre be S(a, 3 – a)

Equation of circle $(x - a)^2 + (y - 3 + a)^2 = 2(1 + a^2)$

$\angle MPN$ reaches maximum value when the circle through M, N and P will be tangent to the x-axis at P

So, $2(1 + a^2) = (a - 3)^2 \rightarrow a = 1, -7$

So the points are P(1, 0) and P′(–7, 0)

$\angle MPN > \angle MP'N$ then P is (1, 0)

40. A, B, C, D

Sol. Based on basic concepts

41. B, C, D

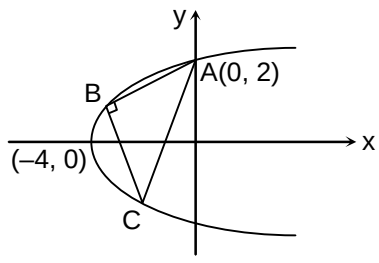
Sol. Suppose $B\left(y_1^2 - 4, y_1\right)$ and $C\left(y^2 - 4, y\right)$

$m_{AB} = \dfrac{1}{y_1 + 2}$

$m_{BC} = -(y_1 + 2)$

So, $y - y_1 = -\left(y_1 + 2\right)\left[y^2 - 4 - \left(y_1^2 - 4\right)\right]$ and $(y \neq y_1)$

We get, $(2 + y_1)(y + y_1) + 1 = 0$

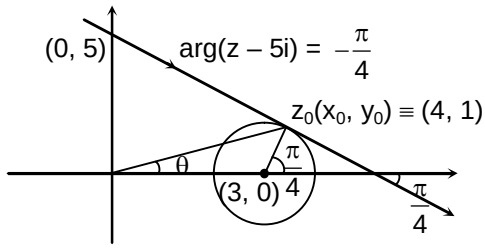


And that is $y_1^2 + (2+y)y_1 + (2y+1) = 0$

For $D \geq 0$ we obtain $y \leq 0$ and $y \geq 4$

42. B, D

Sol. $\Rightarrow \arg(x + i(y-5)) = -\frac{\pi}{4}$



$\Rightarrow \tan^{-1}\left(\frac{y-5}{x}\right) = -\frac{\pi}{4}$

$y - 5 = -x \Rightarrow x + y = 5$

Also, $|z-3| \leq \sqrt{2}$

$\Rightarrow (x-3)^2 + y^2 \leq 2$

Let z_0 is point of contact of line and circle

$\Rightarrow z_0 = 4 + i$

$\tan\theta = \frac{1}{4}$; $\tan 2\theta = \frac{8}{15}$ and $|z_0| = \sqrt{17}$ & $|z_0 - 5i| = 4\sqrt{2}$

43. B

Sol. $S_n - 7\left[\frac{S_n}{7}\right] = 5$

44. A

Sol. Area of $\Delta OCS_1 = \frac{1}{2} \times b \times 6 = 3b$

Semi-perimeter = $\frac{a+b+6}{2}$

Inradius = $\frac{\Delta}{S} = 1 \Rightarrow 5b = 6 + a$

Also, $b^2 + 36 = a^2$

Solving, we get $a = \frac{13}{2}$, $b = \frac{5}{2}$

Maximum distance of normal from centre = $|a - b| = \frac{13-5}{2} = 4$

Length of latus rectum = $\frac{2b^2}{a} = \frac{2 \times \frac{25}{4}}{\frac{13}{2}} = \frac{25}{13}$

$e \Rightarrow \sqrt{\frac{a^2 - b^2}{a^2}} = \frac{12}{13}$

Distance between directrix = $\frac{2a}{e} = \frac{2 \times \frac{13}{2}}{\frac{12}{13}} = \frac{169}{12}$

45. A

Sol. (P) $N_1 = 137$

(Q) $N_2 = 276$

(R) $N_3 = 2008$

(S) $N_4 = 8$

46. B

Sol. (P) $f_1(x) = x^2 \operatorname{sgn}(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$

$f_1'(x) = \begin{cases} 2x & x > 0 \\ 2x & x < 0 \end{cases}$, $f_1'(0) = 0$

$f_1''(x) = \begin{cases} -2 & x < 0 \\ 2 & x > 0 \end{cases}$

So first derivative exists at x = 0 but second derivative does not exist there

(Q) $f_2(x) = \int_0^x t^2 \sin\left(\frac{1}{t}\right) dt = f_2(0) = 0$

$f_2'(x) = x^2 \sin\frac{1}{x}$, $x \neq 0$, $g'(0) = 0$

$f_2''(0) = 0$ but $\lim_{x\to 0} f_2''(x)$ does not exist

(R) $f_3(x) = x^{-\frac{1}{3}} |\sin x|$, $f_3|0| = 0$, $\lim_{x\to 1} f_3(x) = 0$

$f_3'(0^+) = \lim_{x\to 0} \frac{x^{-\frac{1}{3}} \sin x}{x}$ does not exist

(S) $f_4(x) = \begin{cases} 0 & -1 < x \leq 0 \\ -x^3 & 0 < x < 1 \end{cases}$

Second derivative exists and continuous at x = 0

SECTION – D

47. 00002.00

Sol. Average value of –17 and 11 is –3

Let $f(x) = x^3 - 3x^2 + 5x - 3$

$\Rightarrow (x-1)^3 + 2(x-1)$

The graph of f(x) is odd w.r.t to x = 1

The if the root of $x^3 - 3x^2 + 5x - 17 = 0$ is $\alpha = 1 + x_1$

Then the root of $x^3 - 3x^2 + 5x + 11 = 0$ is $\beta = 1 - x_1$

So, $\alpha + \beta = 2$

48. 00003.00

Sol. From $\begin{cases} x + 2y > 0 \\ x - 2y > 0 \\ x^2 - 4y^2 = 4 \end{cases}$

We obtain $\begin{cases} x > 2|y| \geq 0 \\ x^2 - 4y^2 = 4 \end{cases}$

In view of x > 0 and y ≥ 0, we have to find the minimum value of x – y

Let x – y = u substituting into $x^2 - 4y^2 = 4$

We obtain $3y^2 - 2uy + (4 - u^2) = 0$

$\Delta = 4u^2 - 12(4 - u^2) \geq 0$

$\therefore\ u \geq \sqrt{3}$

If $x = \frac{4}{\sqrt{3}}$ and $y = \frac{1}{\sqrt{3}}$, we get $u = \sqrt{3}$

Therefore, minimum value of |x| – |y| = $\sqrt{3}$

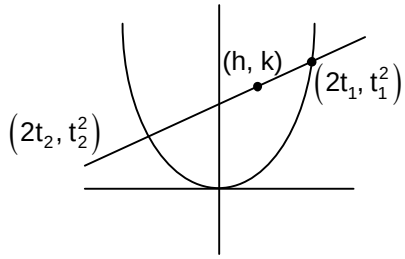
49. 07921.00

Sol. $\tan 1^\circ = \dfrac{\tan(r+1) - \tan r}{1 + \tan r \cdot \tan(r+1)}$

Then $\tan 1^\circ \cdot \tan 2^\circ + \tan 2^\circ \cdot \tan 3^\circ ..... = \cot^2 1^\circ - 89$

50. 00512.00

Sol. $\dfrac{t_1^2 - t_2^2}{2(t_1 - t_2)} = \dfrac{1}{2}$



$\Rightarrow t_1 + t_2 = 1$

Also, $h = \dfrac{4t_1 + 2t_2}{3}$; $k = \dfrac{2t_1^2 + t_2^2}{3}$

$\Rightarrow 3h = 4t_1 + 2(t_2 + 1)$; $3k = 2t_1^2 + (1 - t_1)^2$

$t_1 = \dfrac{3h - 2}{2}$; $3k = 3t_1^2 - 2t_1 + 1$

$3k = 3\left(\dfrac{3h-2}{2}\right)^2 - 2\left(\dfrac{3h-2}{2}\right) + 1$

$\Rightarrow \left(h - \dfrac{8}{9}\right)^2 = \dfrac{4}{9}\left(k - \dfrac{2}{9}\right) \Rightarrow a = 8, b = 2, c = 4$

51. 00001.50

Sol. $1 + x + x^2 ..... = \dfrac{1}{1-x}$ infinite G.P. sum

Differentiate w.r.t. x

$1 + 2x + 3x^2 ..... = \dfrac{1}{(1-x)^2}$

Multiply with x and differentiate, we get

$1 + 2^2x + 3^2x^2 + 4^2x^3 ..... = \dfrac{(x-1)^2 - 2x(x-1)}{(x-1)^4}$

Multiply with x and put $x = \dfrac{1}{3}$, we get $\sum_{k=1}^{\infty} \dfrac{k^2}{3^k} = \dfrac{3}{2}$

So p + q = 5

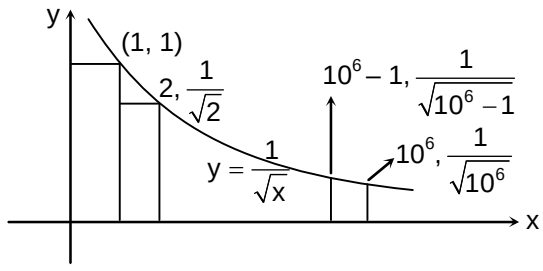
52. 01025.00

Sol. Let x be the number of switches then the minimum number be attempts required can be found by using. Such that if $2^n < x \le 2^{n+1}$

Then n + 1 will be the answer

53. 01998.00

Sol. Consider $y = \dfrac{1}{\sqrt{x}}$



So, $S < \int_1^{10^6} \dfrac{dx}{\sqrt{x}} + 1$

S < 1999

Also, $\int_1^{10^6+1} \dfrac{dx}{\sqrt{x}} < S$

$\Rightarrow$ 1998.001 < S
So, [S] = 1998

54. 01007.00

Sol. So the sum of real coefficients is equal to $S = \frac{1}{2}\left\{(1+i)^{2015} + (1-i)^{2015}\right\} = 2^{1007}$

$\log_2 2^{1007} = 1007 = N$